

Advanced AI Medical Intelligence Platform

Project Technical Evaluation & Architecture Documentation

1. Executive Summary

This project presents an enterprise-grade medical imaging diagnostic solution combining Deep Learning, Explainable AI (Grad-CAM), Database Persistence, and LLM-assisted report generation into a unified web engine.

2. System Architecture & Tech Stack

Component	Technology	Description
Deep Learning	PyTorch / ResNet-18	Pre-trained CNN fine-tuned for classification.
Explainable AI	Grad-CAM	Visual heatmap projection layer targeting feature maps.
Backend API	FastAPI / SQLite	Asynchronous API endpoints with database audit logs.
Frontend UI	Streamlit	User-friendly web dashboard for clinicians.

3. Explainable AI & Clinical Decision Support

To satisfy medical transparency criteria, Grad-CAM overlays highlight key regions contributing to model prediction. An LLM pipeline ingests numerical confidence metrics and classification labels to construct structured diagnostic summaries.

4. Software Quality & Deliverables

The application includes automated unit testing hooks, SQLite audit history tracking, and containerized Docker execution scripts suitable for cloud deployment.